

# Lubrication Degradation Resolved with Oil Service — Cooling Pump A

Asset: AST\_PMP\_001 (Pump) | Bearing: BRG\_005 (SKF6208)  
Fault Mode: lubrication\_issue | ISO Stage: Stage 2  
Learned Case — Knowledge Agent Retrieval Document | DRO Baseline  
CASE\_002 | Outcome: SUCCESSFUL | Opened: 2026-04-18 | Closed: 2026-04-19

## 1. Detection Summary

|                       |                                                               |
|-----------------------|---------------------------------------------------------------|
| Detection Date        | 2026-04-18                                                    |
| Detected By           | monitoring_agent                                              |
| Anomaly Score         | 0.63 (moderate — not critical)                                |
| vib_rms_mm_s          | 4.2 mm/s (baseline mean: 2.5 mm/s — elevated, broadband)      |
| kurtosis              | 3.9 (FT_001 threshold 5.0 — not crossed, fault not confirmed) |
| temp_c                | 75°C (baseline mean: 61°C — rise: 14°C)                       |
| bpfo_energy           | 1.1× (stage_1 threshold 1.3× — sub-threshold)                 |
| bsf_energy            | 1.2× (mild — no dominant ball fault pattern)                  |
| broadband_pattern     | true (all bands rising — no single dominant frequency)        |
| last_lubrication_date | 2026-01-10 — 98 days before detection, overdue by ~38 days    |
| signal_quality_score  | 0.99 (clean signal)                                           |

## 2. Agent Diagnosis

|                   |                                                                                                                                                                                    |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Primary Diagnosis | lubrication_issue — Stage 2                                                                                                                                                        |
| Confidence        | 0.74                                                                                                                                                                               |
| Reasoning         | Broadband vibration increase with temperature rise of 14°C matches FT_003 pattern. No single band dominant. Kurtosis 3.9 below fault-confirmation threshold. Lube overdue 38 days. |
| Differential      | cage_fault early (0.38 confidence) — BSF mildly elevated but no FTF dominance. Lubrication service is correct first action.                                                        |
| Note              | If kurtosis remains above 3.5 after lubrication, escalate to bearing inspection.                                                                                                   |

## 3. Action Taken

|                    |                                                     |
|--------------------|-----------------------------------------------------|
| Recommended Action | Lubrication service (drain and refill), 24h recheck |
| Window Selected    | CTX_002 — LINE_002, 3 hours, approved               |
| Parts Required     | PART_005 (Industrial Oil ISO68, 200 mL, BIN_L2)     |
| Crew Assigned      | K. Mensah (TECH_002, lead)                          |
| LOTO Applied       | LOTO_002 — Valve-7, PRV-02 (is_current: true)       |
| Work Order         | WO_002 — priority MEDIUM                            |

## 4. Findings at Inspection / Teardown

Oil drained from the bearing housing was dark amber with a slightly acrid smell — consistent with oil that has oxidised and lost viscosity from extended service. A light suspension of fine ferrous particles was visible when the sample was held against light, but no gross metallic debris. Quantity insufficient to indicate active bearing surface damage. Bearing housing bore was clean with no scoring. Bearing remained in situ. Oil refilled with 200 mL fresh Industrial Oil ISO68.

5. Outcome

|                               |                                                         |
|-------------------------------|---------------------------------------------------------|
| Post-Service vib_rms (30 min) | 2.8 mm/s (slight improvement — oil not yet distributed) |
| Post-Service vib_rms (24h)    | 2.2 mm/s (QA pass: ≤ 3.5 mm/s) ✓                        |
| Post-Service temp_c (24h)     | 60°C (QA pass: ≤ 65°C) ✓                                |
| Kurtosis at 24h               | 2.8 (below threshold 3.5) ✓                             |
| QA Result                     | PASS — lubrication service resolved condition           |
| Actual Duration               | 2.5 hours                                               |
| Actual Cost                   | INR 4,500                                               |
| Recommendation Followed       | Yes                                                     |

6. Root Cause Confirmed

|                      |                                                                                                                                                                                                    |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Root Cause           | lube_interval_exceeded                                                                                                                                                                             |
| Contributing Factors | Lube re-interval set to 90 days in system but correct interval for this oil type and operating temperature is 60 days. System parameter was incorrect. Updated to 60-day interval after this case. |

7. Lessons Learned

- lubrication\_issue on SKF6208 pump bearings presents as broadband rise + temp rise with kurtosis below 5.0 — reliably distinguishable from outer\_race\_fault.
- First action is always lubrication service and 24h recheck — not bearing replacement. This case resolved fully at ~20% of bearing replacement cost.
- Re-lube interval parameters must be verified against OEM specification for the specific oil type. The system default of 90 days was wrong for this application; correct value is 60 days.
- Vibration improvement after oil service is not immediate — allow 24 hours for fresh oil to distribute. A 30-minute reading after service is for early warning only.
- Post-repair baseline at 24h: 2.2 mm/s vib, 60°C. Expected healthy state for AST\_PMP\_001 with fresh ISO68 oil.